2024 International Conference on Emerging Technologies in Computer Science for Interdisciplinary Applications (ICETCS) | 979-8-3503-7250-2/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICETCS61022.2024.10543936

A Comprehensive exploration of machine learning in early detection with a focus on lung and pancreatic cancer for revolutionizing cancer diagnostics

Ravikumar Ch
Department of AI&DS,
Chaitanya Bharathi Institute of Technology,
Hyderabad, India.
chrk5814@gmail.com

R Ravi Kumar
School of Computer Science & Artificial Intelligence,
SR University,
Warangal, India
ravikumar.racha@gmail.com

Macha Sarada
Department of CSE,
Sri Chaitanya Institute of Technology &Research,
Khammam, India.
saradaramaraok@gmail.com

Mulagundla Sridevi
Department of CSE,
CVR College of Engineering,
Hyderabad, India
sreetech99@gmail.com

Karuna pabba
Department of CSE,
Swami Vivekananda institute of technology,
Hyderabad, India
karunapabba@gmail.com

Md.Aleem Pasha
Department of Mechanical Engineering,
Chaitanya Bharathi Institute of Technology,
Hyderabad, India.
aleemphd2013@gmail.com

***Abstract*-In the ever-evolving landscape of healthcare, this research delves into the integration of machine learning (ML) as a crucial tool for early cancer detection, particularly focusing on lung cancer and pancreatic ductal adenocarcinoma (PDAC). By harnessing electronic medical records (EMR) and innovative urine biomarkers, ML algorithms such as K-Nearest Neighbour, decision trees, Random Forest, and Gradient Boosting classifiers are employed to conduct predictive analysis. The study underscores the critical importance of timely cancer identification in improving patient outcomes and treatment efficacy. Notably, it advocates for novel methodologies to address the pressing clinical needs in PDAC diagnosis, thereby contributing to the ongoing evolution of early cancer detection technologies. Through a comprehensive evaluation of various ML techniques, this research offers nuanced insights into their strengths and limitations, serving as a foundation for advancing healthcare diagnostics. Ultimately, this study holds promise in reshaping the approach to cancer diagnostics and treatment, potentially leading to significant improvements in patient care and outcomes.**

Keywords: Machine Learning, Artificial Intelligence, Pancreatic Cancer, Decision Tree, Gradient Boosting, Clinical Diagnosis, Electronic Medical Records

I. Introduction

Pancreatic cancer, marked by its alarming mortality rate, is a significant public health concern, particularly within gastrointestinal cancers. It stands as the seventh leading cause of cancer-related deaths globally, with a poor 5-year survival rate. Projected to become the second leading cause of cancer deaths by 2030, pancreatic ductal adenocarcinoma (PDAC) faces challenges in early detection due to the lack of reliable screening tools and the anatomical complexities of the pancreas. The asymptomatic nature of the disease further complicates diagnosis, and current imaging techniques often fall short. Over 90% of cases are identified as PDAC, emphasizing the urgent need for early diagnosis and improved treatment modalities. Serum and urinary biomarkers, including Creatinine, LYVE1, REG1A, and TFF1, show promise for non-invasive detection. Efforts are underway to identify suitable classification algorithms to enhance diagnostic accuracy [1].

Meanwhile, lung cancer persists as the foremost cause of tumour-related mortality globally, with its incidence seemingly on a steady rise. Predominantly linked to tobacco use and exposure to asbestos, radon, uranium, and arsenic, lung cancer's high mortality is due to its propensity to metastasize to critical organs. Acknowledging the imperative of early detection and prognosis, machine learning algorithms emerge as pivotal tools in lung cancer diagnosis. Leveraging medical records, these algorithms offer a simplified and understandable model, significantly reducing the risk of misdiagnosis [2]. Addressing the limitations of current diagnostic methods, particularly chest X-rays, machine learning advancements demonstrate their potential to forecast lung cancer cases accurately.

Given cancer's status as the deadliest disease globally, vigilance for early detection is crucial. Lung cancer's insidious progression underscores the need for advancements in diagnostic technologies. Innovations like Low Dose Spiral Computed Tomography (LDCT) mark significant strides in enabling early cancer diagnosis. The survival rate for lung cancer patients significantly improves when detected during the nascent stage, emphasizing the vital role of timely identification facilitated by advanced imaging techniques [3]. Research outcomes, as exemplified by the National Lung Screening Trial (NLST) and NELSON, affirm the efficacy of LDCT imaging in recognizing individuals at high risk, thereby maximizing the chances of early cancer diagnosis [4, 5].

A. Motivations

a) ML's Role in Healthcare: Machine Learning (ML) is a powerful tool in healthcare, particularly for data-driven insights.

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY SILCHAR. Downloaded on January 08,2026 at 17:09:09 UTC from IEEE Xplore. Restrictions apply.

b) Target: Early Lung Cancer Detection: The research focuses on ML for early detection of lung cancer, using decision trees and K-Nearest Neighbor algorithms on electronic medical records.

c) Criticality of Early Lung Cancer Identification: Early recognition of lung cancer is crucial for effective treatment and higher survival rates, where ML plays a pivotal role.

d) Addressing PDAC Challenges: The study tackles challenges in early diagnosis of Pancreatic Ductal Adenocarcinoma (PDAC), proposing ML algorithms for predictive analysis.

e) ML's Versatility in PDAC Prediction: ML algorithms are suggested for PDAC prediction based on urinary biomarkers, showcasing ML's adaptability across different cancer types.

f) Comparison of Model Performance: A key aspect involves comparing classification models, highlighting the effectiveness of Random Forest and gradient-boosting classifiers.

g) Transformative Potential in Cancer Diagnostics: The findings emphasize ML's potential to revolutionize cancer diagnostics, particularly in meeting clinical needs for pancreatic cancer diagnosis.

h) Urgent Need for Innovation: The study underscores the urgency for innovative techniques in pancreatic cancer diagnosis, with ML as a key solution.

i) Insights for Early Cancer Detection: By evaluating different ML methodologies, the research aims to contribute insights for advancing early cancer detection strategies.

j) Overall Impact and Contribution: The study seeks to make a meaningful impact, leveraging ML to enhance cancer diagnostics and improve patient outcomes.

This research navigates the intersection of ML and healthcare, focusing on addressing challenges in PDAC diagnosis. The comparative analysis of ML algorithms reveals their promise in revolutionizing cancer diagnostics, shedding light on the urgency for innovative solutions in the realm of pancreatic cancer diagnosis. With a dual emphasis on the benefits and drawbacks of various ML methodologies, the study strives to shape the landscape of early cancer detection strategies, offering a significant contribution to the ongoing pursuit of improved patient outcomes.

II. Literature review

Machine Learning stands as a significant milestone in AI programming, fostering intelligent learning within systems through the assimilation of prior knowledge and extrapolations from data. This technological leap enables intricate decision-making processes that evolve and learn from past experiences. In the realm of lung cancer detection, various studies have harnessed the power of machine learning algorithms, contributing valuable insights to the advancement of AI-driven solutions in medical diagnostics.

One notable study by Bankar et al. [6] explores the transformative potential of machine learning in healthcare, particularly in lung cancer diagnosis. The researchers employ tree-based algorithms to analyze symptoms across different age groups, identifying key contributors to lung cancer instances. This underscores the role of ML in uncovering data patterns for more accurate diagnoses.

Hussein et al. [7] focus on improving the precision of malignancy-risk determination from radiological images using computer-aided diagnostic (CAD) technologies. Their work utilizes deep learning algorithms, showcasing notable advances in supervised learning and introducing a graph-regularized sparse multi-task learning framework. The study also addresses the challenge of limited labeled training data in medical imaging, exploring unsupervised learning techniques for tumour classification.

Addressing clinical challenges stemming from false negatives in lung cancer diagnosis, Nuhic et al. [8] apply machine learning algorithms to detect benign and malignant pleural effusions without invasive procedures. This innovative approach reduces diagnostic unreliability by leveraging machine learning models to classify lung cancer based on tumour markers from pleural fluids and serum.

Puneet et al. [9] employ machine-learning techniques to enhance the prediction accuracy of lung cancer using frequent blood markers. The study identifies XGBoost with GridSearchCV as the most successful classification model, showcasing the application of machine learning in leveraging regular blood signs for more accurate cancer prediction.

Azzawi et al. [10] delve into the diverse subtypes of lung cancer, emphasizing the importance of accurate classification for targeted therapies and improved patient survival. Their novel approach, Structural Binary Classification (SBC), using the Gene Expression Programming (GEP) algorithm, outperforms other binary decomposition algorithms, providing a valuable resource for precise lung cancer categorization based on tumour structures. Collectively, these studies underscore the transformative potential of machine learning in advancing the field of lung cancer diagnosis and treatment.

A. Contributions

a) This survey focuses on Machine Learning (ML) as a crucial element of Artificial Intelligence (AI) in healthcare data science.

b) Emphasizes the application of ML for early detection of lung cancer, using decision trees and K-Nearest Neighbor algorithms on electronic medical records (EMR).

c) Highlights the survey's outcomes, underscoring ML's transformative potential in revolutionizing cancer diagnostics, with a specific focus on Random Forest and Gradient Boosting classifiers.

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY CALICUT. Downloaded on January 08,2026 at 17:09:09 UTC from IEEE Xplore. Restrictions apply.

III. Methodology

In this comprehensive study, a diverse array of well-established classification techniques was employed, encompassing logistic regression, naive Bayes, decision trees, random forests, K-nearest neighbour (KNN), support vector machines (SVM), and gradient boosting. Rigorous evaluation using performance measures facilitated the identification of the Random Forest Classifier as the most efficient model, achieving an impressive accuracy of 85.59%. The optimization of prediction accuracy was a focal point, achieved through the meticulous selection of pertinent features utilizing Chi-squared feature selection. Subsequently, a meticulous process of feature scaling and data preparation ensued, ensuring a refined and appropriately formatted dataset for input into the classification model.

The classification model, having been fed the transformed dataset, demonstrated its efficacy in generating the intended results. The logical sequence of steps in this study is aptly depicted in the suggested flowchart, illustrated in Fig. 1. This visual representation guides researchers through the systematic progression from feature selection and data preparation to model evaluation, offering a clear roadmap for understanding and replicating the methodology. This multifaceted approach ensures not only the selection of the most robust classifier but also provides insights into the critical steps that contribute to the success of the overall classification framework.

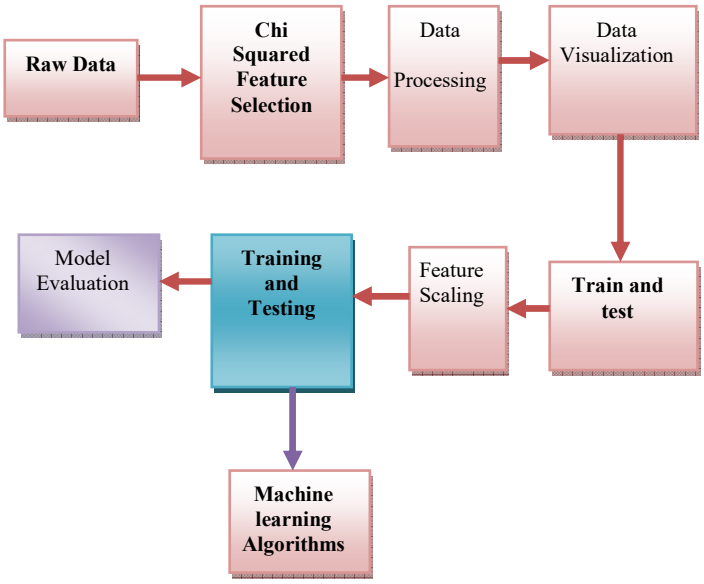

Figure 1: Work Flow Diagram

a) Data Set Source:

Pancreatic cancer was diagnosed using urine biomarkers, namely Creatinine, LYVE1, REG1B, and TFF1. Two hundred-eight people with chronic pancreatitis and other non-cancerous pancreatic illnesses, 199 patients with pancreatic ductal adenocarcinoma (PDAC), and 183 healthy persons had samples obtained. The distribution of data points in each class is displayed in Fig. 2. The original dataset has thirteen input features and one output feature [5].

b) Feature Selection:

Feature selection reduces the dimensionality of the model by identifying pertinent characteristics from the whole dataset and eliminating inappropriate ones. Appropriate characteristics were selected using the Chi-squared test for feature selection. The following features have been chosen from the dataset: REG1A mg/ml, TFF1 mg/ml, LYVE1 mg/ml, Plasma CA19-9 U/ml, Patient's Cohort, Sample Origin, Age, Sex, and so on.

c) Data Preprocessing:

A small number of missing values in specific columns are present in the dataset. It was addressed by applying the linear interpolation approach to substitute missing values for all numerical variables. Elements with more than 50% missing values were not included.

d) **Forecasting Methodologies:**

Depending on the type of sample, the classifier returned an output of 1, 2, or 3 after receiving the input data. The prediction method's output shows which class the input belongs to 1 represents healthy individuals, 2 represents people with pancreatic disorders other than cancer, like chronic pancreatitis, and 3 represents people with PDAC.

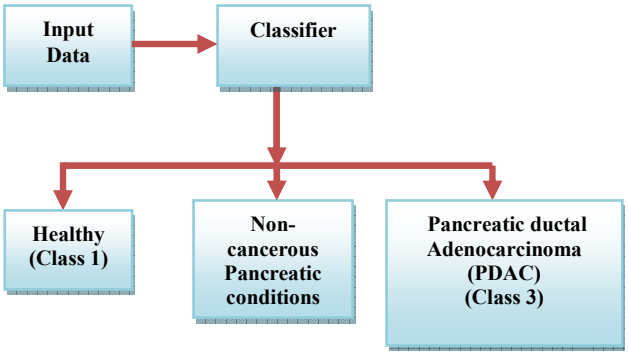

Figure 2: Classification model

This study utilized eight classification algorithms, each playing a role in investigating the features of the dataset (Fig. 2). Key algorithms include:

a) Logistic Regression: A supervised learning algorithm ideal for binary outcome prediction, demonstrating proficiency in distinguishing between healthy individuals, those with non-cancerous pancreatic conditions, and patients with pancreatic ductal adenocarcinoma [6].

b) K-nearest neighbour (KNN): A non-parametric, lazy learning algorithm for classification tasks, classifying new data points based on similarity to labeled data in proximity [7].

c) Support Vector Machine (SVM): Utilizing structural risk minimization, SVM aims to segregate classes by finding an optimal hyperplane with a maximal margin [8].

d) Decision Tree Classifier: Utilizing a tree-like structure, recursively partitioning datasets and minimizing overfitting through tree pruning, offering robustness and efficiency [10, 11].

e) Random Forest: An ensemble of decision trees, collectively providing more accurate outputs through majority voting, particularly effective with larger datasets [12, 13].

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY CALICUT. Downloaded on January 08,2026 at 17:09:09 UTC from IEEE Xplore. Restrictions apply.

f) Gradient Boost: A boosting technique sequentially adding new models to improve the overall accuracy, highly customizable but susceptible to overfitting, mitigated by sub-sampling, shrinkage, and early stopping [14, 15].

g) Multilayer Perception (MLP): A neural network-based classifier relying on interconnected artificial neurons and activation functions [16].

h) Gaussian Naïve Bayes: Leveraging Bayes's theorem assumes independence and equal contribution of features, with continuous values following a normal distribution [17][18].

For model training, 80% of the data was used, and testing employed the remaining 20%, assigning labels 1, 2, or 3 to the testing data.

IV. Evaluation

In this study, multiple classifiers underwent rigorous evaluation to assess their effectiveness in pancreatic cancer classification. Comprehensive metrics, including accuracy, precision, recall, and F1-score, were meticulously computed. The performance details of each classifier are encapsulated in the confusion matrix, outlined in Table 1.

With the rapid evolution of modern medicine, a wide array of data types, including imaging data from X-rays, CT scans, and MRIs, pathology data from tissue sectioning, and molecular data from genomics, transcriptomics, and proteomics, have emerged. Managing this diverse data landscape poses a challenge for clinicians. Traditional clinical information is now complemented by extensive molecular data generated by sequencing technologies. However, patients typically undergo multiple tests, resulting in a complex interplay of various data types. Hence, it's imperative to establish clear methodologies for handling and integrating these diverse datasets effectively. By elucidating strategies for individual data types and exploring methods for integration, clinicians can enhance computational models, providing deeper insights into patient health and disease states.

Accuracy, a comprehensive measure considering both positive and negative instances, quantifies the proportion of true cases among all instances retrieved by the classifiers. Precision, a crucial metric, indicates the ratio of accurately classified positive samples to the total samples labeled as positive, regardless of overall accuracy. In contrast, recall measures the proportion of correctly identified positive samples relative to the total count of positive events, offering insight into the classifiers' performance in pancreatic cancer classification. These metrics collectively provide a nuanced understanding of the classifiers' efficacy in accurately identifying pancreatic cancer cases.

Table 1: Confusion Matrix

	Actual Positive	***Actual Negative***
Predicted Positive	True Positives	False Positives
Predicted Negative	False Negatives	True Negatives

The formulas for the performance measures derived from the confusion matrix are as follows:

$$Accuracy=TP+TNTP+TN+FP+FN \quad (1)$$

$$Precision=TP/TP+FP \quad (2)$$

$$Recall=TP/TP+FN \quad (3)$$

$$F1-score=2\times recall\times precisionrecall+precision \quad (4)$$

$$Specificity=TN/TN+FP \quad (5)$$

During these computations, a crucial focus is placed on the category of healthy patients. True positives (TP) play a pivotal role, representing instances where samples are correctly identified as belonging to the healthy category. On the flip side, false positives (FP) arise when individuals who are actually unhealthy are inaccurately classified as healthy. The scope extends to false negatives (FN), encapsulating situations where healthy individuals are erroneously labeled as unhealthy. Completing the quartet, true negatives (TN) highlight instances where unhealthy individuals are accurately classified as such. This categorization framework, including TP, FP, FN, and TN, serves as a fundamental foundation for evaluating the performance and accuracy of machine learning algorithms in health-related classifications [19].

These calculations consider the category of healthy patients. Instances correctly identified as healthy are referred to as true positives (TP), while instances incorrectly labeled as unhealthy are termed false positives (FP). Instances accurately classified as healthy are labeled as false negatives (FN), and instances correctly identified as unhealthy are designated as true negatives (TN) [20].

Table 2: Precision values

Classifier	***Overall***	***Healthy***	***Non-PDAC***	***PDAC***
Logistic Regression	0.78	0.63	0.79	0.95
Naive Bayes classifier	0.69	0.75	0.59	0.95
KNN	0.76	0.61	0.77	0.92.
Random Forest Classifier	0.86	0.70	0.86	0.92
SVM	0.74	0.62	0.75	0.90
Decision Tree	0.765	0.68	0.70	0.89
MLP Classifier	0.78	0.72	0.77	0.97
Gradient Boosting classifier	0.85	0.73	0.85	0.98

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY CALICUT. Downloaded on January 08,2026 at 17:09:09 UTC from IEEE Xplore. Restrictions apply.

Table 3: Accuracy classification

S.No	*Classifier*	*Accuracy*
1	Logistic Regression	77.3%
2	K-Nearest Neighbor (KNN)	73.77%
3	Support Vector Machine (SVM	73.66%.
4	Decision Tree	77.42%.
5	Random Forest Classifier	86.60%.
6	MLP Classifier	78.02%.
7	Gradient Boosting Classifier|	85.60%
8	Naive Bayes Classifier	65.54%

Table 3 illustrates the accuracy classification of various classifiers. The Random Forest Classifier stands out with the highest accuracy at 86.60%, followed closely by the Gradient Boosting Classifier at 85.60%. Conversely, the Naive Bayes Classifier exhibits the lowest accuracy at 65.54%. Notably, the Decision Tree and Logistic

Regression outperforms the K-nearest neighbour and Support Vector Machine. Further insights reveal the superiority of Artificial Neural Networks over Logistic Regression. Random Forest excels in training sets, while Support Vector Machine performs well in predicting relapse after a year. The K-Nearest Neighbor model dominates in the 2-year relapse risk prediction, surpassing Random Forest and SVM [21][22].

V. Conclusion and Future scope

The application of machine learning techniques holds promising potential for enhancing early diagnosis, leading to improved treatment outcomes and increased patient survival. The careful analysis of performance metrics revealed that the random forest classifier surpassed other considered classifiers, exhibiting the highest accuracy of 86.60%. Both the random forest and gradient boosting classifiers demonstrated comparable performance, emphasizing their effectiveness. The random forest classifier showcased impressive recall, precision, and F1 scores, indicating its suitability for non-invasive pancreatic cancer screening. However, the study underscores the necessity for expanding the dataset to enhance performance further.

The future scope of this research involves addressing current limitations and advancing the model's capabilities. To bolster the model's accuracy, efforts should focus on acquiring more extensive datasets, exploring novel features, and incorporating advanced machine learning methodologies. Continuous research endeavours should aim to refine and broaden the application of non-invasive techniques for early pancreatic cancer screening. The ongoing evolution of machine learning approaches, coupled with a commitment to comprehensive data gathering, will play a pivotal role in achieving significant strides toward accurately identifying and diagnosing pancreatic cancer.

VI. References

[1] Bray, F., Ferlay, J., Soerjomataram, I., Siegel, R. L., Torre, L. A., & Jemal, A. (2018). Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA: A Cancer Journal for Clinicians, 68(6), 394–424. doi:10.3322/caac.21492.

[2] Park, W., Chawla, A., & O'Reilly, E. M. (2021). Pancreatic Cancer: A Review. JAMA, 326(9), 851-862. doi:10.1001/jama.2021.9958

[3] Wong, M. C. S., Jiang, J. Y., Liang, M., Fang, Y., Yeung, M. S., & Sung, J. J. Y. (2017). Global temporal patterns of pancreatic cancer and association with socioeconomic development. Scientific Reports, 7(1), 3165. Doi: 10.1038/s41598-017-03432-8.

[4] Radon, T. P., Massat, N. J., Jones, R., Alrawashdeh, W., Dumartin, L., Ennis, D., et al. (2015). Identification of a Three-Biomarker Panel in Urine for Early Detection of Pancreatic Adenocarcinoma. Clinical Cancer Research, 21(15), 3512-3521. doi:10.1158/1078-0432.CCR-14-3299.

[5] Debernardi, S., O'Brien, H., Algahmdi, A. S., Malats, N., Stewart, G. D., Plješa-Ercegovac, M., et al. (2020). A combination of urinary biomarker panel and PancRISK score for earlier detection of pancreatic cancer: A case-control study. PLoS Medicine, 17(12), e1003489. doi:10.1371/journal.pmed.1003489.

[6] Bankar, A., Padamwar, K., & Jahagirdar, A. (2020). Symptom Analysis Using a Machine Learning Approach for Early Stage Lung Cancer. In 2020 3rd International Conference on Intelligent Sustainable Systems (ICISS) (pp. 246-250). doi:10.1109/ICISS49785.2020.9315904.

[7] Hussein, S., Kandel, P., Bolan, C. W., Wallace, M. B., & Bagci, U. (2019). Lung and Pancreatic Tumor Characterization in the Deep Learning Era: Novel Supervised and Unsupervised Learning Approaches. IEEE Transactions on Medical Imaging, 38(8), 1777-1787. doi:10.1109/TMI.2019.2894349

[8] Nuhic, J., & Kevric, J. (2020). Lung cancer typology classification based on biochemical markers using machine learning techniques. In 2020 43rd International Convention on Information, Communication and Electronic Technology (MIPRO)(pp. 292-297). doi:10.23919/MIPRO48935.2020.9245114

[9] Puneet, & Chauhan, A. (2020). Detection of Lung Cancer Using Machine Learning Techniques Based on Routine Blood Indices. In 2020 IEEE International Conference for Innovation in Technology (INOCON) (pp. 1-6). doi:10.1109/INOCON50539.2020.9298407.

[10] H. Azzawi, J. Hou, R. Alnnni and Y. Xiang, "SBC: A New Strategy for Multiclass Lung Cancer Classification Based on Tumour Structural Information and Microarray Data," 2018 IEEE/ACIS 17th International Conference on Computer and Information Science (ICIS), 2018, pp.68-73, doi: 10.1109/ICIS.2018.8466448.

[11] Arya, K. S., Kavitha, M. N., Abirami, D., & Vaishnavi, D. (2019). Lung cancer detection using machine learning. International Journal of Engineering Research and Technology, 8(11).

[12] Cott J. Adams, Peter Mikhael, Jeremy Wohlwend, Regina Barzilay, Lecia V. Sequist, Florian J. Fintelmann, Artificial Intelligence and Machine Learning in Lung Cancer Screening, Thoracic Surgery Clinics, Volume 33, Issue 4,2023, Pages 401-409.

[13] Hameed, B.S.; Krishnan, U.M. Artificial Intelligence-Driven Diagnosis of Pancreatic Cancer. Cancers 2022, 14, 5382.

[14] Elbanna, K.Y.; Jang, H.-J.; Kim, T.K. Imaging Diagnosis and Staging of Pancreatic Ductal Adenocarcinoma: A Comprehensive Review. Insights Imaging 2020, 11, 58.

[15] Qiao, Z.; Ge, J.; He, W.; Xu, X.; He, J. Artificial Intelligence Algorithm-Based Computerized Tomography Image Features Combined with Serum Tumor Markers for Diagnosis of Pancreatic Cancer. Comput. Math. Methods Med. 2022, 2022, 8979404.

[16] Kaur, R., & Ada, (2013). A study of detection of Lung Cancer using Data Mining Classification techniques. IRJAET, March.

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY CALICUT. Downloaded on January 08,2026 at 17:09:09 UTC from IEEE Xplore. Restrictions apply.

[17] NLST research team, Adams, A. M., Berg, C. D., Clapp, J. D., Fagerstrorm, R. M., Green, I. F., ... Aberle, D. R. (2011). Reduced Lung Cancer Mortality with Low-Dose Computed Tomography Screening, August.

[18] Bunn, B. (2018). Lung Cancer Screening with Low Dose Computed Tomography.

[19] Gurpreet Singh and M. Sachan, "Offline Gurmukhi script recognition using knowledge-based approach & Multi-Layered Perceptron neural network," 2015 International Conference on Signal Processing, Computing and Control (ISPCC), 2015, September, pp. 266-271

[20] Dalianis, Hercules, "Evaluation metrics and evaluation," Clinical text mining, 2018, pp. 45-53.

[21] R. . CH, I. . Batra, and A. . Malik, "A Novel Design to Minimise the Energy Consumption and Node Traversing in Blockchain Over Cloud Using Ensemble Cuckoo Model", IJRITCC, vol. 10, no. 1s, pp. 254–264, Dec. 2022.

[22] Li, Xiawei & Yang, Litao & Zheping, Yuan & Lou, Jianyao & Fan, Yiqun & Shi et al., "Multi-institutional development and external validation of machine learning-based models to predict relapse risk of pancreatic ductal adenocarcinoma after radical resection," Journal of translational medicine, 2021, 19(1), 1-10.

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY CALICUT. Downloaded on January 08,2026 at 17:09:09 UTC from IEEE Xplore. Restrictions apply.